Equipment Operation & Maintenance Manual

Combined Reference for Biomedical Technicians

MediAssist Health Network
Biomedical Engineering Department

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Access: Technicians & Admin

This combined operation and maintenance manual covers the four most common equipment categories deployed across MediAssist hospitals. It is intended for biomedical technicians and trained clinical users. Always observe the device's own labelling and the manufacturer's instructions for use where they are more specific than this manual.

Conventions
Asset tags follow the format **EQ-CAMPUS-XXXX** (e.g. EQ-HYD-1042). Fault codes logged here map directly to the **fault_code** field in the maintenance ticketing system.

A. Patient Monitoring System — MediAssist BM-500

Multi-parameter monitor for ECG, SpO_2, NIBP, respiratory rate and temperature. 12-inch colour display; internal battery backup of 4 hours. Serial-number format: **BM500-YYYY-NNNNN**.

Setup procedure

1. Power on and allow the self-test to complete (~45 seconds).
2. Enter the patient ID.
3. Connect leads and probes in order: ECG leads first, then the SpO_2 probe, then the BP cuff.
4. Confirm a valid trace on each parameter before leaving the bedside.

Alarm parameter defaults and adjustable ranges

Parameter	Default Low	Default High	Adjustable Range
SpO_2	90%	100%	80–100%
Heart Rate	50 bpm	120 bpm	30–250 bpm
NIBP Systolic	90 mmHg	160 mmHg	60–240 mmHg
Temperature	35.5 °C	38.5 °C	34–42 °C

Fault codes

Code	Meaning	Action
E-01	ECG lead off	Re-attach / replace lead
E-02	SpO_2 probe disconnected	Reconnect or replace probe
E-03	NIBP cuff leak	Check cuff and tubing; replace cuff
E-07	Battery low (< 15%)	Connect to mains; service battery if recurrent
E-12	Internal sensor failure	Remove from service immediately; log in CMMS

Maintenance

- **Daily:** clean display with a 70% IPA wipe; verify alarm audibility; check battery charge.
- **Monthly:** SpO_2 verification against a reference pulse oximeter.
- **6-monthly:** NIBP calibration by the biomedical team (tolerance ±3 mmHg).

Remove from service
Fault **E-12** indicates an internal sensor failure — the unit must be removed from service immediately and a maintenance ticket raised.

B. Infusion Pump — DriveFlow IP-200

Syringe and volumetric infusion pump with Dose Rate Error Reduction (DERR) software and a drug library of 150 pre-programmed protocols. Serial-number format: **IP200-YYYY-NNNNN**.

Programming steps

1. Select the infusion mode (syringe or volumetric).
2. Enter the drug name from the on-board library.
3. Enter the patient weight; the system calculates the safe dose range.
4. Confirm the rate against the displayed limits.
5. Start the infusion and verify the running display.

Occlusion pressure alarm settings

- **Default** — 300 mmHg.
- **Venous lines** — reduce the threshold to 200 mmHg.
- **Arterial lines** — maintain the threshold at 300 mmHg.

High-alert drug protocols (drug library)

The following carry **hard limits** (cannot be overridden) and **soft limits** (require confirmation): Dopamine, Dobutamine, Noradrenaline, Insulin, Heparin, Morphine.

Fault codes

Code	Meaning	Action
F-01	Air in line	Purge line; check connections
F-03	Occlusion	Check for kinks/clamps; restart
F-05	Door open	Close door securely
F-08	Battery failure	Service battery; use mains
F-12	Drug library update required	Remove from service; update library; log in CMMS

Maintenance schedule

- **After each patient use:** clean with 70% IPA.
- **Weekly:** functional check by the biomedical team.
- **Annual:** full service by an authorised service engineer.
- **Every 18 months:** replace the battery.

C. Autoclave Steriliser — SterilPro 3000

134-litre chamber steam steriliser supporting gravity and pre-vacuum cycles, validated for wrapped instruments, unwrapped instruments and porous loads. Serial-number format: **SP3000-YYYY-NNNNN**.

Cycle selection

Cycle	Temperature	Pressure	Hold Time	Suitable For
Gravity 121	121 °C	103 kPa	15 min	Unwrapped instruments, fluids
Gravity 134	134 °C	206 kPa	3 min	Unwrapped solid instruments
Pre-vacuum 134	134 °C	206 kPa	3.5 min	Wrapped sets, porous & hollow loads

Routine testing

- **Bowie-Dick test:** mandatory every morning before the first load. A fail means **do not use** — call the biomedical team.
- **Biological indicator (BI) test:** weekly. A positive result means quarantine all loads from that week and notify infection control.

Daily log requirements

Record for every cycle: cycle number, cycle type, temperature/pressure chart, pass/fail result, operator ID, and load contents.

Fault codes

Code	Meaning	Action
E-01	Door seal failure	Inspect/replace gasket; leak test
E-04	Steam generator low water	Refill / check supply
E-07	Temperature sensor fault	Do not use; call biomedical
E-11	Vacuum pump failure	Do not run pre-vacuum cycle; service pump

D. Portable X-Ray Unit — RadiPro MX-150

150 kW mobile X-ray unit with a wireless DR panel, DICOM compatibility and battery operation for ~50 exposures per charge. Serial-number format: **MX150-YYYY-NNNNN**.

Technique chart

Body Part	kVp	mAs	Grid	Notes
Chest AP (adult)	90	5	No	Erect preferred
Chest AP (paediatric)	70	3	No	Short exposure time
Abdomen AP	80	15	Yes	Expiration phase
Pelvis	85	20	Yes	Gonad shielding
Knee AP	65	8	No	—

Radiation safety

- All personnel stand **> 2 m** from the unit, or behind a protective barrier, during exposure.
- The operator must wear a dosimetry badge at all times.
- Log the exposure count weekly.

Fault codes

Code	Meaning	Action
F-02	DR panel not detected	Re-pair panel; check battery
F-05	Battery below 20%	Charge before use
F-09	kV generator fault	Remove from service; log in CMMS

Battery & preventive maintenance

- Charge after each session; do not store fully discharged; replace every 3 years.
- **Monthly:** clean, battery check, exposure-counter log.
- **Quarterly:** HV cable inspection, collimator check.
- **Annual:** kV/mAs calibration by a service engineer, plus a radiation-leakage test.

> **Remove from service**
> Fault **F-09** (kV generator fault) requires the unit to be removed from service and logged in the CMMS. Do not attempt further exposures.

E. Maintenance & Fault-Code Summary

Quick cross-reference of the fault codes that mandate immediate removal from service. These are the codes that should appear on escalated maintenance tickets.

Code	Device	Meaning	Mandatory Action
E-12	BM-500 Monitor	Internal sensor failure	Remove from service; raise ticket
F-12	DriveFlow IP-200	Drug library update required	Remove from service; raise ticket
F-09	RadiPro MX-150	kV generator fault	Remove from service; raise ticket

F. Preventive Maintenance Calendar (summary)

Equipment	Daily	Weekly	Monthly	Annual
BM-500 Monitor	Clean, battery	—	SpO_2 check; NIBP 6-monthly	Full service
DriveFlow IP-200	Clean per use	Functional check	—	Service + 18-mo battery
SterilPro 3000	Bowie-Dick	BI test	Seal inspection	Validation & calibration
RadiPro MX-150	—	Exposure log	Clean, battery	kV/mAs calibration; leakage test

Record-keeping
All preventive maintenance and faults must be recorded in the CMMS / maintenance ticketing system. Escalated tickets are reviewed weekly by the Biomedical Engineering lead.

G. Commissioning & Acceptance Testing

Every new or repaired device must pass commissioning before clinical use. The biomedical team records the result against the asset tag in the CMMS, checking that:

- **Electrical safety** — earth leakage is within IEC 60601 limits.
- **Functional test** — all parameters read within manufacturer tolerance.
- **Calibration certificate** — is valid and on file.
- **Asset tagging** — the EQ-CAMPUS-XXXX label is applied and registered.
- **User training** — ward staff are trained and the sign-off is recorded.

H. Decontamination Before Service

Mandatory
No device may be handed to biomedical engineering or an external service engineer until it has been decontaminated and a **decontamination certificate** attached. This protects service staff from exposure.

- Surface-clean per the device section above (70% IPA unless contraindicated).
- Visibly soiled equipment must be cleaned and, where applicable, disinfected before transport.
- Attach the completed decontamination certificate to the device and the ticket.

I. Common Operator Errors & Troubleshooting

Symptom	Likely Cause	First Check
No SpO_2 reading	Probe placement / perfusion	Reposition probe; warm the digit
NIBP repeatedly retries	Cuff size / patient movement	Use correct cuff size; keep limb still
Infusion not starting	Door not latched / clamp closed	Close door; open the roller clamp
Autoclave cycle aborts	Overloaded chamber / wet load	Reduce load; ensure correct packing
X-ray panel image blank	Panel not paired / low battery	Re-pair DR panel; charge battery

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